

ANANNYA POPAT

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EDUCATION

University of Toronto

Master's, Applied Computing, AI Concentration

Sep 2023 - Jan 2025

GPA: 3.94

PROFESSIONAL EXPERIENCE

Magical

AI Forward Deployment Engineer

May 2026 - Present

Toronto, ON, Canada

- Built production-grade multimodal, multi-agent AI workflows to automate end-to-end Prior Authorization (PA) and Eligibility & Benefits (EBV) processes across browser-based and desktop healthcare applications.
- Designed intelligent automation pipelines that extracted and validated patient information from EHR/EMR systems, then accurately populated payer portals to streamline Prior Authorization submissions and Eligibility & Benefits verification.

NextPathway

AI Engineer (Automation Lead)

Aug 2025 - Apr 2026

Toronto, ON, Canada

- Analyzed client requirements and migration code across diverse ETL and data platforms, including Informatica, SQL, Spark, and DataStage for migration to DBT, SnowSQL, Dataproc, ADF/Databricks, etc., along with visual presentation using PowerBI and Excel.
- Designed and implemented LLM-powered multi-agent pipelines using LangGraph and OpenAI APIs to optimize crawl and translation for legacy codebase to Snowflake/GCP-powered systems with 90% accuracy - boosting migration performance by 30% and reducing manual effort by 50%.
- Developed a synthetic data generation engine to create test data for evaluating the performance of translated/migrated code-bases.
- Enabled the organization to attain Snowflake's highest partner designation and drive multi-million-dollar revenue growth across enterprise client engagements.

University Health Network (UHN)

Machine Learning Specialist (Team Lead)

Feb 2025 - Aug 2025

Toronto, ON, Canada

- Engineered an interactive web-based 3D anatomical model from CT scans using MedSAM2, TotalSegmentator, VTK and GenAI to enhance surgical planning and improve patient outcomes. Utilized on-prem NVIDIA GPUs to fine-tune models and run experiments.
- Integrated fine-tuned GramGAN model into backend via Docker containers to apply realistic medical textures onto anatomical structures.
- Implemented a Tool-Tissue Interaction (TTI) detection model using YOLOv11 (transfer learning) and fine-tuned Vision Transformer (ViT) model for precise tool and interaction classification in the OR to advance surgical practice in real-time.
- Fine-tuned ResNet50 model to classify histologic types in early gastric cancer (EGC) from endoscopic images, achieving 91% specificity for undifferentiated types and 87% specificity for differentiated type

University Health Network (UHN)

AI Research Intern

May 2024 - Dec 2024

Toronto, ON, Canada

- Led the development of an interactive and realistic 3D anatomical model from patient-specific CT scans to enhance surgical planning.
- Leveraged nnU-Net based TotalSegmentator for precise tissue segmentation and implemented 3D modeling using Visualization Toolkit (VTK) for interactive rendering.
- Performed a comparative study on texture extraction and mapping for medical images using Wavelet Transform, Gabor Filters, fine-tuned Neural 3D Style Transfer, and fine-tuned GramGAN, where deep learning techniques performed 61.4% better than traditional methods.

AdGlobal360

Data Science Intern

May 2022 - Jul 2022

Gurugram, HR, India

- Developed an lead scoring prediction model using XGBoost to identify potential buyers based on website activity with 95% F1 score.
- Worked with big data and performed exploratory data curation and analysis using SQL, Spark and Python frameworks for accessing client data entries for ML modeling.

KEY PROJECTS

(AnvayaAI) Explainer: Research paper to Animated Videos

Founder and Lead Engineer (Personal Project)

Toronto, ON, Canada

- Designed an agentic AI system that transforms research papers into intuitive visual explanation videos using manim engine
- Implemented multi-agent architecture for explanation generation, LLM (Gemini 3.1 Pro) judge and manim code generation.
- Paired explanation generation agent with LLM-judge to evaluate content on 6 heuristics, designed after iterative prompt engineering.
- Paired code generation agent with execution-based Python and Manim renderer to ensure the code is runnable.
- Implemented failure-recovery module for code generator by pairing with RAG over Manim documentation using ChromaDB for grounding.
- Integrated beat-level OpenAI TTS narration and dynamically synchronized animation timing with audio, adjusting render speeds to ensure precise audiovisual alignment.
- [Link to project](#)

Optimized LLM and RAG Modeling: Classification and Instruction Fine-Tuning

Personal Project

Toronto, ON, Canada

- Integrated attention mechanisms like causal, multi-query, and grouped multi-query attention with KV cache optimization.
- Experimented with positional embedding strategies in Transformers such as sinusoidal, learned, and relative position embeddings.
- Implemented classification and instruction fine-tuning of GPT2 using different prompt styles and data processing techniques.
- Deployed on AWS Sagemaker using an MLOps pipeline for automated training, validation, and deployment.
- Built a RAG agent (Agentic AI) for Pokemon Battle Analysis using Gemini API, LangChain and Streamlit
- [Link to project](#)

Pocket Comedian: BosonAI Hackathon Winner

Toronto, ON, Canada

- Developed a real-time “pocket comedian” application using Audio LLMs, Flask and Socket.IO, enabling seamless speech-to-speech comedic interactions in an engaging UI.
- Integrated the Higgs Audio Understanding Model for speech input processing, Qwen3 for generating context-aware comedic responses, and the Higgs Audio TTS Model for expressive comedic response.
- Implemented features such as response-voice selection based on your favorite comedian, different comedic modes, multi-turn conversation memory for better banter and integration of Mandarin and French input and audio responses.
- [Link to project](#)

Ink-To-Tint: Manga Artisan

Master's Academic Project, University of Toronto

Toronto, ON, Canada

- Automated manga colorization and style conversion to enhance readability and ease artists' workload.
- Optimized image processing techniques like dodging and dilation to decolorize colored manga datasets.
- Trained a Pix2Pix conditional GAN from scratch with quantization to colorize manga pages with a 55% decrease in MSE loss.
- Applied knowledge distillation using pre-trained Stable Diffusion model (MeinaMix v10) to style transfer across four distinct manga styles.
- [Link to project](#)

Text-based 3D Gaussian Splatting Object Segmentation

Master's Academic Project, University of Toronto

Toronto, ON, Canada

- Developed a 3D Gaussian Splatting segmentation model using LangSAM for multi-model text-driven 3D segmentation.
- Devised an optimized prompt initialization strategy employing K-means clustering for optimal view selection and point sampling.
- Improved IoU by 3%, accuracy by 1%, and reduced computational load by 50% while maintaining ear-optimal results.
- Improved results for the paper "SAGD: Boundary-Enhanced Segment Anything in 3D Gaussian via Gaussian Decomposition".
- [Link to project](#)

InstantSplat Implementation: Unbounded Sparse-view Pose-free Gaussian Splatting

University of Toronto

Toronto, ON, Canada

- Implemented key concepts from the InstantSplat paper preceding official code release to enable optimized 3D Gaussian Splatting.
- Replaced traditional SfM (COLMAP) pipelines with a transformer-based pose estimation method inspired by DUST3R.
- Reconstructed 3D scenes efficiently using joint optimization of camera extrinsics and gaussians and fewer input images under 40 seconds!
- [Link to project](#)

Qualitative Badminton Player Detection and Analysis

Vellore, TN, India

- Detected the exact boundary points of the badminton court using Probabilistic HoughTransform and K-means Clustering.
- Tracked near and far-side badminton players in videos using the Particle Filter algorithm with 99% accuracy.
- Predicted badminton strokes (net-drop return, defense, smash) of near-side player using Pillow and CNNs, with 81% accuracy, with the aid of pose estimation algorithm using OpenPos.
- [Link to project](#)

SKILLS

Skills: Python, R, Java, SQL, HTML/CSS, Pytorch, LangChain, LangGraph, Tensorflow, OpenCV, Scikit-learn, Streamlit, Flask, Blender, Git, Linux/Unix, Docker, Computer Vision, 3D Computer Vision, Large Language Models, Deep Learning, Machine Learning, Data Engineering, Natural Language Processing (NLP), Google Cloud Platform, AWS Sagemaker, Microsoft Azure, Snowflake, Apache Kafka, Kubernetes, NumPy, Pandas, C/C++, VTK, NiBabel, Spark, REST APIs, NLTK, Hadoop, React.js, Node.js, PowerBI, XML, JSON